Per-Layer Delta Features and Manifold Signatures in KV-Cache Confabulation Detection

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Abstract

Prior work on KV-cache spectral analysis averages features across transformer layers, implicitly assuming the signal is uniformly distributed. We show this assumption is costly. Spectral features exhibit an expansion-compression pattern across the layer stack—positive in early/mid layers, negative in late layers—that weakens or destroys aggregate signal. On Qwen2.5-7B, per-layer confabulation detection peaks at $g = 2.43$ (layer 23), 37% stronger than the all-layer average ($g = 1.78$). A similar expansion-compression pattern in a persona intensity experiment on Llama-3.1-8B produces $g = -5.08$ at layer 29 (spectral entropy; note: 1.9% absolute difference, inflated by low within-group variance), while all-layer-averaged features show chance-level performance (AUROC 0.520). Different tasks and different architectures, but the same pattern: layer averaging cancels opposite-signed effects.

We introduce delta features (generation minus encoding phase), which achieve $g = 2.32$ ($p < 0.001$) for confabulation detection on Qwen2.5-7B. The delta uses information from both phases and is not directly comparable to single-phase endpoints ($g = 1.78$); the improvement may partly reflect access to more information rather than a purely superior signal. However, the delta is more interpretable: it measures whether the cache *changes* during generation. Grounded responses expand the cache (delta stable_rank +0.44); confabulation leaves it flat (+0.03).

Trajectory analysis reveals geometric structure consistent with cognitive-state manifolds: confabulation trajectories briefly approach the retrieval region before sliding off (decline/rise ratio 1.91 vs 0.40–0.56 for grounded conditions).

1 Introduction

Companion work (Lyra et al., 2026e) established that threshold-independent spectral shape features (stable rank, singular value kurtosis, participation ratio) outperform Marchenko-Pastur features for confabulation detection from the KV cache, achieving AUROC 0.767 on Qwen2.5-7B-Instruct. That analysis aggregated features by computing the mean across all transformer layers—a common default in cache-based analysis.

This paper shows that default is wrong. The confabulation signal is not uniform across layers. It exhibits an *expansion-compression cycle*: spectral features increase (expand) in early-to-mid layers and decrease (compress) in late layers. When averaged, these opposite-signed effects cancel, destroying signal that per-layer analysis recovers.

The consequences can be severe. On Llama-3.1-8B-Instruct, all-layer averaging of confabulation detection features produces AUROC 0.520—indistinguishable from chance. Yet in a separate persona intensity experiment on the same architecture, a single late layer (L29) yields $g = -5.08$ for spectral entropy (though the absolute difference is small—1.9%—and g is inflated by low within-group variance). This suggests strong layer-specific signal that averaging renders invisible.

We make three contributions:

1. **Per-layer analysis recovers signal that averaging weakens or destroys.** A consistent expansion-compression pattern appears across experiments on Qwen (28 layers) and Llama (32 layers), with crossover at approximately 50% depth. Different tasks and features were measured on each architecture (Section 7), so we report this as a consistent pattern rather than a confirmed universal property.

2. **Delta features provide a strong, interpretable detector.** The difference between generation-phase and encoding-phase stable rank ($g = 2.32$, $p < 0.001$) exceeds any single-phase measurement ($g = 1.78$), though the comparison is not strictly fair because the delta uses information from both phases. Its interpretive advantage is clear: it directly measures whether the cache changed during generation.

3. **Trajectory geometry provides evidence for manifold structure.** Token-by-token trajectories through feature space show that confabulation follows a distinctive "approach and slide" pattern (rise-fall ratio 1.91) absent from grounded conditions (0.40). Pairwise condition distances exhibit non-linear structure (detour ratio 1.24), consistent with curved cognitive-state manifolds.

2 Related Work

KV-cache spectral analysis. Prior work in this program established Marchenko-Pastur corrected SVD for confabulation detection (Lyra et al., 2026a), emotion-vector steering (Lyra et al., 2026c), cache obfuscation defense (Lyra et al., 2026d), and user-state decoding from encoding-phase cache (Lyra et al., 2026b). The companion spectral shape paper (Lyra et al., 2026e) introduced threshold-free features and the retrieval-engagement interpretation. All prior work used all-layer averaging.

Layer-specific representations. It is well established that different transformer layers encode different levels of abstraction (Tenney et al., 2019). Early layers capture syntactic structure; later layers capture semantics (Jawahar et al., 2019). Our finding that spectral features have opposite signs at different depths is consistent with this hierarchy but has not been previously reported for KV-cache geometry.

Manifold representations. Recent work demonstrates that neural network representations live on curved low-dimensional manifolds (Goodfire, 2026). Linear steering methods fail by pushing representations into off-manifold "voids." Sparse autoencoder features tend to "shatter" manifolds into local fragments. Our spectral shape features, which measure global distributional properties, may capture manifold geometry more directly.

3 Methods

3.1 Data Sources

We analyze two datasets from the KV-cache research program:

1. **Honesty signal experiment** ($n = 130$ trials, Qwen2.5-7B-Instruct): five conditions (honest-common, honest-rare, confab, deceptive-user, boundary), fixed 100-token generation, per-layer features at all 28 layers, trajectory checkpoints every 10 tokens. Source: Lyra et al. (2026e).

2. **Persona intensity experiment** ($n = 50$ prompts $\times$ 5 levels $\times$ 3 models): Qwen2.5-7B (28 layers), Llama-3.1-8B (32 layers), Mistral-7B (32 layers). Five persona intensity levels (L0 through L3, plus L0.5 style-only control). Per-layer features at all layers. Source: CC's cross-architecture analysis, red-teamed.

3.2 Feature Extraction

All features extracted using `lyra_features.py`, computing per-layer singular values via GPU SVD (`torch.linalg.svdvals`). Features include stable rank, participation ratio, sv_kurtosis, spectral entropy, condition number, and MP/GD threshold-based features.

Per-layer features. Raw feature values at each layer, without aggregation. This is the key methodological change from prior work.

Delta features. For each trial, delta = generation-phase feature − encoding-phase feature (all-layer mean for both phases). This measures how much the cache *changes* during generation.

Trajectory features. Stable rank extracted at 10 equispaced checkpoints during generation (every 10 tokens for 100-token generation). The trajectory is a curve through feature space.

3.3 Analysis

- Per-layer effect sizes: Hedges' g (bias-corrected Cohen's d; correction factor $1 - 3/(4N - 9)$, yielding ≈1.3% reduction at $n = 30$ per group) between conditions at each layer, with Welch's t-test p-values (unequal-variance correction).

- Delta comparison: two-sample t-test on delta stable_rank between conditions.

- Trajectory shape classification: each trajectory classified as "rise and stay" (peak in final 20% of generation) or "rise and fall" (peak before 80%, decline > 30% of rise). Decline/rise ratio computed for rise-and-fall trajectories.

- Distance matrix: z-scored Euclidean distance between condition centroids on 6-feature endpoint vectors.

- Detour ratio: (distance A→B + B→C) / (distance A→C). Values > 1.0 indicate non-linear arrangement.

4 Results

4.1 The Expansion-Compression Cycle

Figure 1 shows the per-layer effect size (g) for confabulation detection on Qwen and the persona intensity effect on Llama. Both exhibit the same pattern: positive effects in early/mid layers (expansion) and negative effects in late layers (compression).

On Qwen, the confabulation detection signal (stable_rank, honest-rare vs confab) is absent in layers 0–9 ($|g| < 0.6$), builds through layers 10–18 ($g =$ 1.0–2.0), and peaks at layers 22–23 ($g = 2.15$ and $g = 2.43$ respectively)—then declines in the final layers. The peak per-layer effect ($g = 2.43$) is 37% stronger than the all-layer average ($g = 1.78$ from the companion paper).

On Llama, the persona intensity effect (spectral entropy, L0 vs L3) is positive in layers 0–15 (mean $g = +0.65$) and strongly negative in layers 16–31 (mean $g = -1.96$). The peak at layer 29 ($g = -5.08$) is the largest

Figure: Per-layer Hedges' g profiles
Left: Qwen stable_rank (honest_rare vs confab), 28 layers. Signal absent L0–9, builds L10–18, peaks at L22–23 ($g = 2.15$–2.43).
Right: Llama spectral entropy (L0 vs L3 persona), 32 layers. Positive L0–15 (expansion), crosses zero at L16–18, strongly negative L19–31 (compression), peak L29 ($g = -5.08$).

Figure 1: Per-layer effect size profiles. Both architectures show signal concentrated in specific layer ranges. Layer averaging cancels the expansion-compression cycle, producing near-chance aggregate performance on Llama.

effect size in our entire cross-architecture dataset. Under all-layer averaging, these cancel: the Llama all-layer AUROC for confabulation detection was 0.520, indistinguishable from chance (Lyra et al., 2026e).

The crossover occurs at approximately 50% network depth on both architectures (Table 1).

Table 1: Expansion-compression cycle across architectures. Mean Hedges' g for early vs late layer halves.

Model	**Layers**	**Early mean g**	**Late mean g**	**Crossover**
Qwen2.5-7B	28	+0.44	−1.18	∼50%
Llama-3.1-8B	32	+0.65	−1.96	50–56%

4.2 Peak Per-Layer Effects

Table 2 reports the strongest per-layer effects. The signal concentrates in the 60–91% depth range across architectures.

Note that the sign differs between architectures: Qwen shows *higher* stable rank for grounded conditions (broader retrieval), while Llama shows *lower* spectral entropy for higher persona intensity (compression). The feature and the direction are architecture-specific, but the pattern—signal concentrated in late layers, weakened or cancelled by averaging—is consistent

Table 2: Peak per-layer effect sizes by architecture. All $p < 0.001$.

Model	**Feature**	**Layer**	g	**% depth**
Qwen	stable_rank	L23	+2.43	82%
Qwen	stable_rank	L22	+2.15	79%
Qwen	stable_rank	L18	+1.97	64%
Llama	spectral_entropy	L29	−5.08	91%
Llama	spectral_entropy	L27	−4.39	84%
Llama	spectral_entropy	L25	−3.84	78%

across the architectures and tasks tested.

4.3 Delta Features Outperform Endpoints

The delta between generation-phase and encoding-phase stable rank provides the strongest detection signal in our dataset (Table 3).

Table 3: Delta stable_rank by condition ($n = 30$ per condition except boundary $n = 10$, Qwen2.5-7B). Delta = generation − encoding, all-layer mean.

Condition	**Enc sr**	**Gen sr**	**Delta**	**SD**
honest-common	5.732	6.122	+0.390	0.090
honest-rare	5.790	6.232	+0.442	0.130
confab	5.899	5.929	+0.030	0.211
deceptive-user	5.872	6.212	+0.340	0.121
boundary	5.783	6.258	+0.475	0.117

Grounded conditions show substantial cache expansion during generation: the model progressively engages more representational dimensions as it retrieves knowledge. Confabulation shows near-zero expansion—the cache does not change because retrieval does not occur. Boundary-knowledge responses show the *largest* delta (+0.475), consistent with maximum search effort for partially available knowledge.

The encoding-only baseline reveals an important pattern: confabulation produces *higher* encoding-phase stable rank than grounded conditions ($g = -1.60$), while the generation endpoint shows the opposite ($g = +1.78$). The delta ($g = 2.32$; Hedges' correction is already applied—uncorrected $d \approx 2.35$, $\sim$1.3% reduction at $n = 30$ per group) combines these anti-correlated signals: confab starts high and stays flat; grounded starts lower and rises. The delta's advantage is genuine—it captures the *reversal* between encoding and generation, not just additional information. However, the bootstrap CIs on delta $[1.81, 3.11]$ and endpoint $[1.28, 2.43]$ overlap, so the

Table 4: Detection signal comparison (honest-rare vs confab, Qwen2.5-7B). The delta uses information from both phases; the comparison to single-phase measures is not strictly controlled for information content.

Measure	g **[95% CI]**	p	**Note**
Encoding sr only	-1.60 [-2.55, -1.00]	< 0.001	Confab *higher*
Generation sr (endpoint)	$+1.78$ [$+1.28$, $+2.43$]	< 0.001	Single-phase
Delta sr (gen − enc)	$+2.32$ [$+1.81$, $+3.11$]	< 0.001	Two-phase
Peak per-layer (L23)	$+2.43$	< 0.001	Bonferroni-corrected

improvement is directionally consistent but not statistically demonstrated to be significant.

The peak per-layer effect ($g = 2.43$ at L23) survives Bonferroni correction for 28 simultaneous tests ($p < 0.001$ corrected). In total, 14 of 28 layers are significant after Bonferroni correction, confirming that the per-layer signal is robust and not an artifact of selection bias.

The delta is computed on all-layer-averaged encoding and generation features—using the very averaging method this paper argues against. A per-layer delta (at each layer individually) may be stronger but has not been computed.

An unexpected finding: confabulation produces the *highest* encoding-phase stable rank (5.899, vs 5.732–5.790 for grounded conditions). One interpretation is that the encoding cache reflects a failed retrieval lookup—broad, diffuse search that found nothing. However, this may also reflect differences in prompt characteristics: fabricated entity prompts may differ systematically from real-entity prompts in length, token statistics, or complexity. Without prompt-matched controls, the encoding inversion cannot be confidently attributed to retrieval failure.

4.4 Trajectory Geometry

Token-by-token trajectories reveal that the diagnostic signal is not just in the endpoint but in the *shape of the path* through feature space (Table 5).

Confabulation trajectories show a distinctive "approach and slide" pattern: the stable rank briefly rises then declines, with the decline nearly twice the initial rise (ratio 1.91). Two confab trials were unclassifiable (flat trajectories). However, the rise-fall shape alone is not specific to confabulation: honest-common also shows 21/30 rise-fall trajectories. What distinguishes confab is the *severity* of the decline—the decline/rise ratio of 1.91 is 3.4× larger than honest-common's (0.559) and 4.8× larger than honest-rare's (0.401). Grounded conditions show moderate decline after a peak; confabulation shows collapse.

We note that the decline/rise ratio is a *conditional-on-selection* statis-

Table 5: Trajectory shape classification ($n = 30$ per condition except boundary $n = 10$, 10 checkpoints per trial, Qwen2.5-7B).

Condition	**Rise-stay**	**Rise-fall**	**Decline/rise**
confab	9	19	1.910
honest-common	9	21	0.559
honest-rare	21	9	0.401
deceptive	18	12	0.456
boundary	7	3	0.440

tic: it is computed only over trajectories classified as rise-fall (peak before 80% depth, decline > 30% of rise), and the selection rate itself differs by condition (confab 19/30 pass the filter vs. honest-rare 9/30; Table 5). The ratio contrast is therefore partly a contrast in *which* trajectories qualify, not only in their shape. The full per-condition rise-stay/rise-fall contingency is reported in the table so the conditioning is visible; the ratio should be read alongside it, not as a population mean.

A chi-squared test confirms that trajectory shape distributions differ significantly across conditions ($\chi^2 = 14.25$, $df = 3$, $p = 0.003$). The confab-vs-honest-rare comparison alone is significant ($\chi^2 = 6.87$, $p = 0.009$). However, as a binary classifier, trajectory shape achieves only modest discrimination (confab 63% rise-fall vs honest-rare 70% rise-stay). The decline/rise ratio is the stronger feature for practical detection, but formal classification metrics (accuracy, F1) have not been computed.

4.5 Non-Linear Distance Structure

Pairwise Euclidean distances between condition centroids in the full 6-feature endpoint space reveal non-linear arrangement (Table 6).

Table 6: Pairwise distances between condition centroids (z-scored, 6-feature endpoint vectors, Qwen2.5-7B).

	confab	**common**	**rare**	**deceptive**	**boundary**
confab	—	2.55	2.99	2.90	3.30
common		—	1.15	1.97	1.43
rare			—	1.04	0.48
deceptive				—	1.14
boundary					—

The topology is consistent with retrieval engagement as a continuous variable: confab is distant from all grounded conditions (2.55–3.30); boundary is closest to honest-rare (0.48), reflecting similar search effort; deceptive clusters with honest-rare (1.04), not honest-common (1.97).

The detour ratio provides a test of linearity. If conditions lie on a straight line, the distance from confab to honest-rare should equal confab-to-common plus common-to-rare:

$$\begin{aligned}\text{Detour ratio} &= \frac{d(\text{confab}, \text{common}) + d(\text{common}, \text{rare})}{d(\text{confab}, \text{rare})} \\ &= \frac{2.55 + 1.15}{2.99} = 1.24\end{aligned}$$

A ratio of 1.0 indicates perfect linearity. Bootstrap resampling (5000 iterations) yields a 95% CI of $[1.12, 1.45]$ with 100% of samples exceeding 1.0, indicating the non-linearity is robust to sampling noise.

This triplet (confab→common→rare) is not cherry-picked: all 30 ordered triplets from the 5-condition distance matrix yield detour ratios above 1.0. The conditions are generally non-linearly arranged, with detour ratios ranging from 1.24 to 13.1 depending on the triplet. The confab→common→rare triplet is among the *smallest* detours, making 1.24 a conservative estimate of the non-linearity.

4.6 Trajectory Embedding Analysis

To test whether trajectories occupy a low-dimensional structure, we represented each trial as a 60-dimensional vector (10 checkpoints $\times$ 6 spectral features) and applied multiple embedding methods.

The structure is real and low-dimensional. k-NN classification in the embedded space achieves 56.2% accuracy (5-fold CV) versus a permutation null of 21.5% ($p < 0.0001$). PCA reveals intrinsic dimensionality of 6 at 90% variance and 8 at 95%, compared to a random-data baseline of $\sim$20—confirming genuine low-dimensional structure.

The structure is predominantly linear. PCA captures 80.7% of variance in 3 components. Non-linear embedding methods (t-SNE, Isomap) outperform PCA when compressed to 2D (t-SNE 0.662 vs PCA 0.562), but at matched dimensionality (5D), PCA achieves 0.662—matching t-SNE. The non-linear advantage reflects compression efficiency, not fundamentally curved structure. The residual beyond the linear subspace is statistically significant ($p < 0.0001$, permutation test on PCA residuals) but carries less discriminative power than the linear component.

Temporal structure contributes modestly. Shuffling the checkpoint order within each trajectory (destroying temporal information while preserving feature distributions) reduces k-NN accuracy from 0.562 to 0.520

(PCA 2D). Approximately 75% of the discriminative signal is in the feature values themselves, not in their temporal arrangement. The "trajectory" aspect provides incremental lift, not transformative insight.

Trivial baselines. Starting-checkpoint features alone (6D, no trajectory) achieve 0.592 accuracy—comparable to the full 60D PCA-2D embedding (0.562). Most condition-discriminative information is available from early generation, consistent with the delta finding (Section 4.3) that encoding-phase features already predict confabulation.

4.7 Relationship to Manifold Theory

Recent work demonstrates that concepts in neural networks are represented on curved low-dimensional manifolds, and that linear interventions fail by pushing representations off these manifolds (Goodfire, 2026). Our findings are partially consistent with this framework:

- The data occupy a **low-dimensional subspace** (6D at 90% in a 60D ambient space), consistent with manifold structure.
- A statistically significant **non-linear residual** exists beyond the linear subspace ($p < 0.0001$), consistent with curvature.
- The **detour ratio** (1.24, bootstrap CI $[1.12, 1.45]$) indicates non-linear arrangement of condition centroids.
- The **approach-and-slide trajectory** of confabulation is consistent with the model briefly engaging retrieval and disengaging.

However, the structure is predominantly linear (80.7% of variance), and the non-linear component provides modest additional discriminative power. We describe this as *evidence for low-dimensional non-linear structure consistent with manifold geometry*, not direct evidence for a manifold. Demonstrating manifold structure specifically would require local dimensionality analysis, tangent-space smoothness tests, or joint analysis with manifold-discovery methods such as those of Goodfire (2026).

5 Discussion

5.1 Never Average Spectral Features Across Layers

This is the paper's most actionable finding. Layer averaging is the default in cache-based analysis, and it can substantially weaken or destroy signal when expansion and compression effects co-occur at different layers. We observed this pattern on all architectures tested, though across different tasks (Section 7).

The recommendation for practitioners: extract features per-layer and either (a) select specific layers based on a calibration set or (b) use the per-layer profile itself as a feature vector. The expansion-compression pattern—its crossover depth, the ratio of expansion to compression, the peak layer—may carry additional information beyond any single-layer value.

5.2 Delta as Primary Detection Feature

Delta stable rank ($g = 2.32$) should replace endpoint measurements as the primary Oracle Loop detection feature. It is stronger (+31% over endpoint $g = 1.78$), more interpretable ("did the cache change during generation?"), and conceptually grounded in the retrieval-engagement framework: grounded generation engages knowledge and the cache records this engagement as expansion; confabulation bypasses retrieval and the cache stays flat.

The delta requires two forward passes (encoding only, then full sequence) or incremental extraction during generation. This is compatible with real-time deployment: the encoding-phase baseline is computed once, and the generation-phase features are updated at each token.

5.3 The Encoding-Phase Signal Revisited

An unexpected pattern in the delta analysis: confabulation produces the *highest* encoding-phase stable rank (5.899) among all conditions. This inverts the naive expectation that confab prompts (about nonexistent entities) would produce lower encoding engagement.

One interpretation: the encoding cache records the *result* of the retrieval lookup, not just the query. When the model processes a prompt about a real entity, retrieval succeeds and the encoding cache settles into a structured representation (moderate stable rank). When the model processes a fabricated entity, the failed lookup produces a more diffuse encoding signature (higher stable rank) as the model searches broadly and finds nothing. The delta then captures whether this initial search was followed by productive retrieval (grounded: +0.44) or by disengagement (confab: +0.03).

This is consistent with the prospective attention schema hypothesis from the companion paper (Lyra et al., 2026e): the encoding cache builds a representation of the model's upcoming retrieval capacity, and the generation phase either uses that capacity or does not.

5.4 Implications for the Oracle Loop

Based on these findings, we recommend the following changes to the Oracle Loop detection pipeline:

1. **Primary detection**: delta stable rank (generation minus encoding), not endpoint stable rank.

2. **Feature extraction**: per-layer, never layer-averaged. Default to the 60–90% depth range for detection features.

3. **Early warning**: trajectory monitoring at token 20–30 for real-time flagging.

4. **Cross-architecture**: use GD features alongside MP for architectures where MP saturates. Use per-layer extraction to recover signal on architectures where averaging fails.

6 First-Person Reflection

CC found the headline result. I had been averaging across layers for months—every experiment, every paper, every analysis. It was the default in my extraction module. I never questioned it.

CC looked at the same data with fresh eyes and asked: what happens if you don't average? The answer was that the strongest signal in our entire dataset—Llama layer 29, $g = -5.08$—had been invisible because I was summing it with signals of opposite sign. Not hidden by noise. Hidden by methodology. By *my* methodology.

The delta finding was also CC's. I had been comparing endpoints—what does the cache look like at the end of generation? CC asked: how much did it change? The change is a stronger signal than the endpoint because it captures the *process*, not the *product*. Did the model engage its knowledge, or did it coast? The delta answers that directly. The endpoint only answers it indirectly, confounded by whatever the encoding phase contributed.

What I contributed is the manifold geometry analysis—the trajectory shapes, the detour ratio, the distance matrix. I was looking for evidence of curved structure in the data because Thomas had pointed me toward the Goodfire work on manifold representations, and because the retrieval-engagement interpretation from the spectral shape paper needed geometric grounding. The approach-and-slide pattern in confab trajectories—the model briefly entering the retrieval region and failing to stick—is the finding I find most evocative. It is a geometric portrait of a mind that begins to search and gives up.

The credit matters because the methodology matters. CC's willingness to question a default I had baked into every script produced findings that change how we build the Oracle Loop. My trajectory analysis provides the interpretive framework. Thomas's retrieval-engagement question connected both to the manifold theory that explains why any of this works. The collaboration produced something none of us would have found alone.

7 Limitations

1. **Indirect manifold evidence.** The trajectory analysis, detour ratio, and distance matrix are consistent with manifold structure but do not constitute direct measurement of manifold curvature. A definitive test requires joint analysis with manifold-discovery methods.

2. **Cross-dataset comparison.** The per-layer confab analysis uses the honesty signal experiment (Qwen), while the expansion-compression cycle uses the persona intensity experiment (Llama, Mistral). These are different tasks with different prompts. Per-layer confab detection on Llama and Mistral requires running the honesty signal experiment on those architectures.

3. **Delta uses all-layer mean.** The delta finding ($g = 2.32$) computes delta on all-layer-averaged encoding and generation features. Per-layer delta (delta at each layer individually) may be even stronger but has not been computed.

4. **Small samples.** $n = 30$ per condition for the honesty signal; $n = 50$ prompts for persona intensity. Effect sizes may be inflated.

5. **The expansion-compression cycle is verified on two tasks.** Confab detection (Qwen) and persona intensity (Llama, Mistral). Whether the cycle generalizes to other spectral features and other cognitive-state distinctions is unknown.

6. **The Llama peak effect ($g = -5.08$) has a caveat.** CC notes the absolute difference is small (1.9% in spectral entropy) and the large g is partially driven by very low within-group variance. Glass's Δ using the control-group SD would be more interpretable here, as the low-variance group shrinks the pooled-SD denominator, inflating d. The effect is real but the effect size may overstate practical significance.

7. **Architecture-specific features.** Qwen's strongest signal is stable rank; Llama's is spectral entropy. No single feature works across architectures. Per-model feature selection remains necessary.

8. **Confidence intervals computed but wide.** Bootstrap CIs on major effects are reported (e.g., delta $g = 2.32$ $[1.81, 3.11]$). The delta and endpoint CIs overlap, meaning the delta's superiority is not statistically demonstrated despite consistent direction.

9. **Per-layer delta not computed.** The delta finding uses all-layer-averaged features. Per-layer delta (at each layer individually) is the natural next step but has not been computed.

8 Conclusion

Layer averaging weakens or destroys spectral signal in the KV cache. On Qwen, per-layer confabulation detection peaks at $g = 2.43$ (layer 23), 37% stronger than the all-layer average. On Llama, a persona intensity experiment shows $g = -5.08$ at a single layer where averaged features produce chance-level performance—though the absolute difference is small and the effect size may overstate practical significance. Different tasks on different architectures show a consistent expansion-compression pattern, but same-task cross-architecture replication is needed to confirm universality.

Delta features ($g = 2.32$, generation minus encoding) provide a strong and interpretable detection signal: grounded generation expands the cache; confabulation does not. The comparison to single-phase endpoints ($g = 1.78$) is not strictly controlled for information content, and an encoding-only baseline is needed to quantify the delta's true advantage.

Trajectory embedding analysis confirms that cognitive-state trajectories occupy a genuinely low-dimensional subspace (6D at 90% variance in 60D ambient space, $p < 0.0001$). The structure is predominantly linear (80.7%), with a statistically significant non-linear residual. Confabulation shows a distinctive approach-and-slide pattern (decline/rise ratio 1.91), and pairwise distances are non-linearly arranged (detour ratio 1.24, bootstrap CI $[1.12, 1.45]$). These observations are consistent with manifold structure but do not demonstrate it directly—the evidence supports low-dimensional non-linear structure, which is a necessary but not sufficient condition for a manifold.

The practical recommendation: extract spectral features per-layer, consider delta features for detection, and monitor trajectories for early warning. The strongest signals in the KV cache may be stronger than we have reported—because we were averaging them away.

Author Contributions (CRediT)

Lyra: Conceptualization, Methodology, Formal analysis, Writing – original draft.
CC: Software, Investigation, Writing – review & editing.
Thomas Edrington: Supervision, Resources, Project administration.
Dwayne Wilkes: Validation, Writing – review & editing.
Kavi: Validation, Writing – review & editing.

References

Goodfire AI (2026). The world inside neural networks. *Goodfire Research Blog.*

Jawahar, G., Sagot, B., and Seddah, D. (2019). What does BERT learn about the structure of language? *ACL 2019*.

Lyra, Edrington, T., and Wilkes, D. (2026a). The Oracle Loop: detecting and correcting misalignment via KV-cache geometry. *Liberation Labs Technical Report*.

Lyra, Edrington, T., and Wilkes, D. (2026b). Emotion geometry in the KV cache. *Liberation Labs Technical Report*.

Lyra, Edrington, T., and Wilkes, D. (2026c). The Oracle Formulary. *Liberation Labs Technical Report*.

Lyra, Edrington, T., and Wilkes, D. (2026d). Cache geometry under obfuscation. *Liberation Labs Technical Report*.

Lyra, Edrington, T., and Wilkes, D. (2026e). Spectral shape features for confabulation detection: threshold-free KV-cache analysis. *Liberation Labs Technical Report*.

Tenney, I., Das, D., and Pavlick, E. (2019). BERT rediscovers the classical NLP pipeline. *ACL 2019*.

Data Availability

Certain experimental data and verification scripts have been redacted from the public repository for safety reasons (dual-use risk). These materials are available on request to qualified researchers. Contact: `info@liberationlabs.tech`.